Project: Molecular dynamics simulation of solid crystalline phases

Objective: To study the structure of a solid system in different crystalline phases.

Outline

1. Write a MD code for a (N,V,T) simulation of a Lennard-Jones (LJ) system. Use N $\leq$ 500 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.

2. Create two initial set-ups of positions:

   I) from the sites of a bcc (*body centred cubic*) lattice. (HINT: consider a bcc lattice as the superposition of 2 single cubic lattices.)

   II) from the sites of a fcc (*face centred cubic*) lattice. (HINT: consider a fcc lattice as the superposition of 4 single cubic lattices.)

3. For each of the initial set-ups of positions, create an initial set-up of velocities from a Maxwell distribution. Remember that the velocity of the center of mass of the system must be zero.

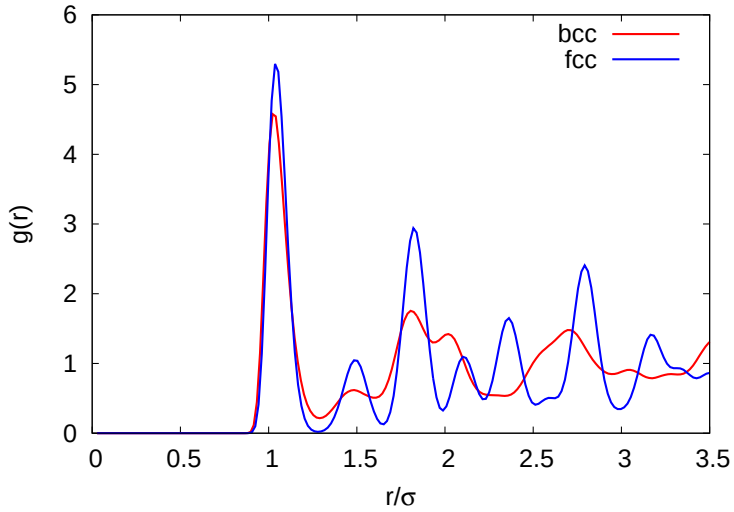
4. For each of the initial set-ups of positions+velocities, run a MD simulation for solid argon at T* = 1.0 and ρ^*= 1.2. Compute the interaction energy U^*, the pressure P^*, the radial distribution functions $g(r)$, and the mean square displacement $r^2(t)$.

5. Repeat the simulations for the following temperatures: T*= 2.0, T*=3.0, T*=4.0 and T*=5.0, and compare with the previous results.

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\AA$)
Ar	39.948	119.8	3.405

Some results to compare with:



Radial distribution functions for the different crystalline phases at T*=1.0 .

Questions

1. **Show the equivalence between the leap-frog Verlet and the velocity Verlet algorithms.**

2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

3. Give an interpretation of the peak positions shown by the $g(r)$ functions at T*=1.0.

4. Discuss for both crystalline phases the differences you find in the $g(r)$ and $r^2(t)$ functions when heating the system.